

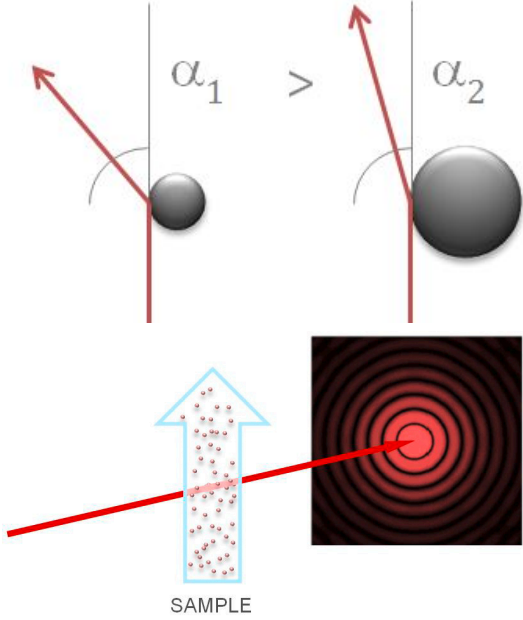
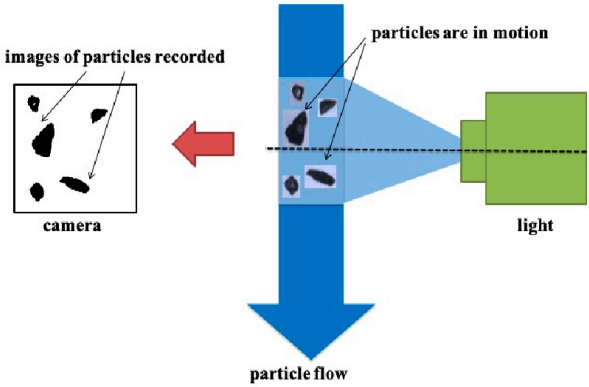
Particle Size Analysis

There are many different techniques for particle size analysis, including:

- Dynamic light scattering
- Laser diffraction
- Sieve analysis
- Sedimentation
- Microfluidic diffusional sizing
- Imaging particle analysis

The technique you choose should depend on the expected particle size range, if the sample is liquid or solid, the amount available, and the application of the sample.

The Breakerspace particle size analyser uses both laser diffraction and dynamic imaging particle analysis (DIA).

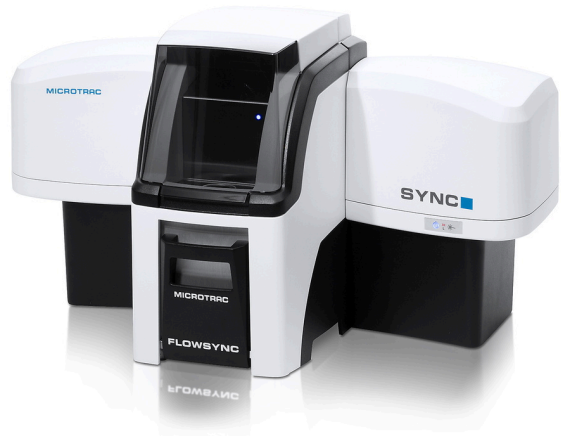
Laser diffraction	Dynamic imaging particle analysis
A laser beam is shot through a sample. The beam diffracts at different angles depending on the particle size. The resulting diffraction pattern is detected and analyzed with an algorithm.  The diagram illustrates the principle of laser diffraction. A laser beam (red line) passes through a sample of particles (represented by a blue arrow labeled 'SAMPLE'). The beam is diffracted at different angles, labeled α_1 and α_2, where $\alpha_1 > \alpha_2$. The resulting diffraction pattern is shown as concentric rings. A camera is positioned to capture the pattern.	The particles in your sample are dispersed, then travel through the measurement zone using gravity, air pressure, or some liquid. A high speed camera takes images of a dispersion of your sample's particles. These images are analyzed to determine the particle size distribution  The diagram illustrates the principle of dynamic imaging particle analysis (DIA). Particles are shown in motion through a measurement zone. A high-speed camera captures images of the particles, which are then analyzed to determine the particle size distribution. The diagram shows a blue arrow labeled 'particle flow' moving downwards through a measurement zone. A green box labeled 'light' is positioned to the right. A red arrow labeled 'images of particles recorded' points to a camera. DIA can offer information about particle size, shape, and velocity.

What are some applications of particle size analysis?

- Analyzing the shape of drug particles in formulations.
- Analyzing cell cultures for cell shape and morphology in biology.
- Measuring the size and shape of cement particles, which affect its surface area, compressive strength and curing time.

Instructions on using the Breakerspace Particle Size Analyzer

The Breakerspace has a Microtrac Sync Particle Size Analyzer, which gives particle size distribution data for dry powder samples with particles between 0.01 and 4,000 micrometers using laser diffraction.



Instrument startup:

- Power on the instrument if needed
- Log on to instrument workstation using your MIT Kerberos
- Open Microtrac FLEX software
- From the Open Analyzer menu select Sync Analyzer
- Wait while the instrument initializes

Operation:

- Check the active database, change/create new if needed
- Load measurement SOP, if no SOP exists ask lab staff for assistance
- Clean loose particles from sample chamber
 - The tray can be removed to be cleaned separately, and debris tends to get trapped between the tray and the slab that it sits on.
- Load sample, 1/4 tsp for small sample tray
 - Keep your sample behind the marker at the front, so that a background test can be run before the sample is processed.
- Run Auto-Sequence
- Repeat as needed

Instrument shut down:

- Export data as needed
- Close the FLEX software
- Log off Windows workstation
- Instrument may be left powered on

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!